WELCO MEDICAL & DENTAL SOLUTIONS • OFFICIAL ENGINEERING REPORT

System Enhancements & Implementation Guide

An executive, non-technical walkthrough of recent major platform upgrades across Customer Journey, Marketplace Quoting, Account Privacy, and Catalog Administration.

TARGET AUDIENCE	PLATFORM SCOPE	FEATURES DELIVERED	STATUS
Stakeholders & Admins	Storefront & Admin Portals	8 Core Upgrades	100% Deployed & Verified

1 Customer Order & Shipment Experience

F-01 Place Order: Unified Consignment Reference Number

COMPLETED

Standardizes order identification so hospital buyers and shipping couriers share the exact same shipment code from checkout to delivery.

✕ PREVIOUS LIMITATION

Order confirmations showed internal database IDs. Customers could not easily find their courier dispatch reference.

✓ IMPLEMENTED SOLUTION

Every placed order automatically generates and highlights a clean **Consignment Reference** (e.g. `WL-2026-1022`) with a 1-click copy button.

HOW IT WORKS IN PRACTICE

Directly on the checkout success screen, the consignment reference is prominently displayed with instant track-order buttons. This reference links seamlessly into courier delivery manifests and support tickets.

F-02 Order Tracking: Frictionless Public Lookup by Consignment Ref

COMPLETED

Enables any buyer or logistic partner holding a consignment reference to track package status without requiring login credentials.

✕ PREVIOUS LIMITATION

Entering numbers with # or extra spaces produced errors. Tracking also required mandatory account login.

✓ IMPLEMENTED SOLUTION

Built smart input sanitization that strips # and spaces automatically, and enabled secure public tracking without sign-in hurdles.

HOW IT WORKS IN PRACTICE

Whether a customer types `#WL-1520`, `wl-1520`, or pastes with accidental spaces, the system normalizes the query instantly, fetching milestones (Confirmed, Processing, Shipped, Delivered) with real-time visual progress bars.

2 Marketplace Quoting & Product Presentation

F-03 Direct Quote: Dynamic Multi-Currency Switching

COMPLETED

Empowers international dental clinics and hospitals to view instrument prices and request formal quotes in their preferred local currency.

✕ PREVIOUS LIMITATION

All product prices and quotes were locked exclusively in US Dollars ($). Clients in UAE, Saudi Arabia, Europe, and Egypt could not change currency.

✓ IMPLEMENTED SOLUTION

Integrated an active currency selector across Product Detail, Cart, and Checkout (USD, EUR, AED, SAR, GBP, EGP) with live conversion rates.

HOW IT WORKS IN PRACTICE

Selecting **AED** converts prices dynamically in real-time. When requesting a Direct Quote, the customer's selected currency is saved and transmitted to sales reps, guaranteeing the formal quotation arrives in the exact currency requested.

F-04 Product Details: Bulleted Technical Specifications List

COMPLETED

Displays essential surgical and orthodontic instrument dimensions directly beneath the product title in a fast, scannable format.

✕ PREVIOUS LIMITATION

Technical specifications were hidden in bottom tabs or formatted as dense paragraphs that were cumbersome for clinicians to scan.

✓ IMPLEMENTED SOLUTION

Placed a clean bulleted list directly below the instrument name, highlighting dimensions, materials, autoclave ratings, and wire limits.

VISIBLE INFORMATION HIERARCHY

- **General Specifications:** Autoclave rating (134°C), German Stainless Steel grade (DIN 1.4021), CE Class IIa certification.
- **Wire Capacity:** Dedicated badge indicating maximum wire gauge (e.g., up to .022" x .028").
- **Wire Cutting Capacity:** Explicit thresholds for hard vs. soft wire cutting without cluttering descriptions.

F-05 Interactive Instrument Inspection: Zoom In, Out & Pan

COMPLETED

Allows surgeons and dental procurement specialists to scrutinize instrument tips, serrations, and hinge alignment in ultra-high resolution.

✕ PREVIOUS LIMITATION

The "Inspect Instrument" popup was completely static with no zoom controls or ability to navigate fine instrument details.

✓ IMPLEMENTED SOLUTION

Built an interactive inspection viewer equipped with Zoom In (+), Zoom Out (-), 100% Reset, mouse wheel zooming, and click-and-drag panning.

HOW IT WORKS IN PRACTICE

Clicking "Inspect Instrument" opens a distraction-free studio lightbox. Clinicians can zoom up to 300% to examine blade edges or serrations with silky-smooth drag navigation.

3 Customer Account Privacy & Multi-Tenancy Security

F-06 Customer Accounts: 100% Strict Data Isolation

COMPLETED

Guarantees that every registered customer strictly accesses their own personal orders, quotes, and communications—never records of other clinics.

✕ SECURITY RISK

Certain account portal queries lacked customer ID locks, risking potential visibility of other clients' commercial proposals and purchase inquiries.

✓ IMPLEMENTED SOLUTION

Enforced strict cryptographic tenant filters at both the database query layer and the front-end user portal components.

PROTECTED ACCOUNT SECTIONS

- **Account Dashboard:** Metric summary cards (Total Orders, Spend, RFQs) calculate exclusively from the customer's own activity.
- **Quote Requests (RFQs):** Only RFQs initiated by the customer or their verified clinic account are displayed.
- **Price Quotes:** Commercial proposals and confidential pricing discounts are restricted solely to the recipient.
- **Order History:** Past purchase receipts and tracking history belong strictly to the logged-in client.
- **Product Inquiries:** Technical consultations and instrument questions are private between the client and Welco support.

4 Admin Management & Catalog Architecture

F-07 Admin Catalog: Explicit "Most Selling Product" Control

COMPLETED

Provides store administrators with direct editorial authority over which instruments appear in the storefront's "Most Selling" carousel.

✕ PREVIOUS LIMITATION

Newly created products automatically entered the "Most Selling" carousel because the query sorted by creation date, confusing site visitors.

✓ IMPLEMENTED SOLUTION

Added a dedicated **"Feature as Most Selling Product"** toggle in the admin portal and updated catalog queries to respect admin curation.

HOW IT WORKS IN PRACTICE

When adding or editing a product, the admin can deliberately toggle the "Most Selling" badge on or off. Standard new items stay neatly in their catalog categories without unintentionally crowding the featured sales section.

F-08 New Product Form: Structured Bullet Specs & Capacities

COMPLETED

Standardizes technical specification entry in the admin portal with dedicated optional fields for orthodontic wire capacities.

DEDICATED ADMIN INPUT FIELDS

- **Specification (Optional):** Multi-line box where each line renders as a clean bullet point on the storefront.
- **Wire Capacity (Optional):** Specific gauge rating (e.g. `.022" x .028"`). If left empty, it hides automatically.
- **Wire Cutting Capacity (Optional):** Maximum wire cut threshold (e.g. `Hard Wire ≤ 0.56 mm`).

5 Catalog Integrity: Category Cascade Deletion

F-09 Category Cleanup: Automatic Cascading Product Deletion

COMPLETED

Maintains database cleanliness and storefront stability by automatically removing all child products whenever an obsolete category is removed.

✕ PREVIOUS RISK

Deleting a category left orphaned products in the database with missing category links, causing broken storefront filters or 404 search errors.

✓ IMPLEMENTED SOLUTION

Implemented automatic cascading removal: deleting a category automatically marks every product assigned to it as deleted in the database.

HOW IT WORKS IN PRACTICE

When an administrator confirms category removal, the database executes an atomic cascade update. All linked products are removed simultaneously, guaranteeing that zero orphaned records remain visible to customers.

✓ Executive Implementation Summary Matrix

FEATURE NAME	DOMAIN	CORE BUSINESS VALUE	STATUS
Consignment Reference	Customer	Clear, standardized shipment reference shared across invoices & couriers	✓ Live
Public Order Tracking	Customer	Frictionless tracking without login; auto-cleans # symbols & spaces	✓ Live
Multi-Currency Quoting	Marketplace	Real-time currency rates with locked quote selection (USD, EUR, AED, SAR, etc.)	✓ Live
Bulleted Specifications	Marketplace	Prominent spec checklist directly beneath instrument title for rapid scanning	✓ Live
Inspection Modal Zoom	Marketplace	Interactive 300% zoom, wheel scroll, and pan navigation for clinical examination	✓ Live
Account Data Isolation	Security	100% privacy: clients only see their own quotes, orders, and inquiries	✓ Live
Most Selling Admin Toggle	Admin	Direct admin curation over featured high-selling instruments on homepage	✓ Live
Product Capacities Form	Admin	Dedicated optional fields for wire & cutting capacities with clean UI fallback	✓ Live
Category Cascade Delete	Admin	Prevents orphaned products by auto-deleting child items with parent category	✓ Live

Data Privacy & Multi-Tenancy
Database-level client isolation protects hospital bids, pricing, and purchase records.

Global Trade Ready
Seamless currency switching accommodates procurement teams across the GCC and Europe.

Zero Catalog Errors
Cascading category deletions eliminate broken links and orphaned database records.